

Lab: Epistemological Analysis of the Free Energy Principle

Learning Goal: Apply philosophical analysis tools to the FEP's epistemological claims.

Part 1: Mapping the Epistemological Tradition

Exercise: Complete the mapping table and evaluate the strength of each analogy:

Philosophical Concept	FEP Analog	Key Similarity	Key Difference	Strength (1-5)
Kant's categories	Generative model	Both structure experience a priori	Kant's are fixed; generative models are learned	
Justified true belief	Model evidence	Both concern epistemic warrant	FEP has no "truth" — only predictive success	
Cartesian doubt	Prior uncertainty	Both start from uncertainty	Descartes seeks certainty; FEP accepts irreducible uncertainty	
Empiricist sensation	Sensory states	Both ground knowledge in experience	Empiricism is passive; AIF is active	
Pragmatist truth	Predictive success	Both define truth through consequences	Pragmatism is social; FEP is individual	
Phenomenological intentionality	Prediction as directedness	Both describe mind-world relation	Husserl brackets ontology; FEP assumes a causal structure	

Write a 200-word analysis of the strongest and weakest mappings. For the weakest, explain specifically what would need to change in the FEP to improve the analogy.

Write your response here

Part 2: The Markov Blanket Debate

Learning Goal: Evaluate competing interpretations of MB ontology.

Exercise: Read the following three positions and argue for each:

Position A (Realist): "Markov Blankets are real statistical structures that exist independently of our analysis. A cell membrane is a Markov Blanket whether or not anyone describes it as such."

Position B (Constructivist): "Markov Blankets are modeling tools imposed by the analyst. The same system can be partitioned differently depending on the question being asked."

Position C (Processual): "Markov Blankets are temporary, dynamically maintained structures. They are neither fixed objects nor mere constructs — they are processes of self-organization that can form, dissolve, and reform."

Write 150 words defending Position A, 150 words defending Position B, and 150 words defending Position C. Finally, state your own position with reasons. Cite at least one philosophical precedent for your position (e.g., Whitehead for processual, Quine for constructivist).

Write your response here

Part 3: Circularity Analysis

Learning Goal: Diagnose and evaluate the self-evidencing circularity.

Exercise:

1. Formalize the circularity: Model M defines what counts as evidence E. E is used to update M. There is no external check.
2. Is this vicious circularity (fatal to the framework) or virtuous circularity (a feature)?
3. Compare with at least two other cases of circularity in philosophy or science:
4. The hermeneutic circle (understanding parts requires understanding the whole, and vice versa)
5. Defining health in terms of function and function in terms of health
6. The measurement problem in quantum mechanics (the observer is part of the system)
7. **New:** Does the FEP's circularity differ from autopoiesis (Maturana & Varela)? Both involve self-maintaining systems. Is one more epistemologically problematic than the other?

Write your response here

Part 4: Is the FEP Falsifiable?

Learning Goal: Apply philosophy of science criteria to the FEP.

Exercise: Apply three philosophy-of-science frameworks:

1. **Popper:** State the FEP as precisely as possible. What observation would *falsify* it? If none, is it a tautology?
2. **Lakatos:** Is the FEP a *research programme* with a "hard core" and "protective belt"? What is the hard core? What auxiliary hypotheses constitute the protective belt? What would constitute a *degenerating* shift?
3. **Kuhn:** Is the FEP a *paradigm* in Kuhn's sense? What would a Kuhnian "crisis" look like for the FEP? What "anomalies" exist?
4. Compare with other unfalsifiable-seeming frameworks (natural selection, thermodynamics, Bayesian probability). Which comparison is most apt?

Write your response here

Part 5: Reflection

In 250 words, reflect on this question: Does the FEP provide genuine epistemological insight, or does it merely translate existing philosophical ideas into mathematical notation?

Specifically:

- Name one insight that the FEP provides that was *not already available* in the philosophical tradition
- Name one philosophical problem that the FEP fails to address despite appearing to
- Does mathematical formalization add genuine understanding, or does it trade philosophical nuance for the illusion of precision?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Historical mapping	FEP in philosophical tradition
2	Dialectical argument	Markov Blanket ontology
3	Circularity analysis	Self-evidencing and autopoiesis
4	Philosophy of science	Falsifiability, research programmes, paradigms
5	Critical reflection	Novelty and limits of the FEP